

From image information to regional strain report

a Image evidence

What is directly visible in cine-CMR

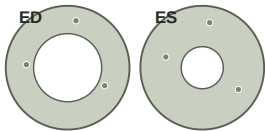


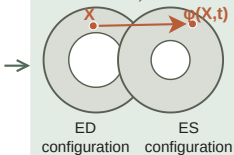
Image features (e.g., boundaries, texture, intensity)

Material labels not directly observed

Observed image information

b Inferred mapping

Which material point corresponds across time (estimated, not observed)



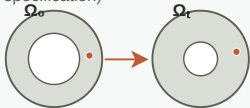
Feature tracking · registration · biomechanics · learning-based

Categories may overlap; each constrains the undetermined correspondence through its own assumptions.

Candidate correspondence is inferred by the estimator and cannot be directly read from ordinary cine images.

c Defined strain

How deformation is quantified (mathematical definition + specification)



$$u(X,t) = \varphi(X,t) - X$$

$$F = \partial\varphi/\partial X = I + \partial u/\partial X$$

$$E = \frac{1}{2}(F^T F - I)$$

Green–Lagrange strain (example)

Specification required:

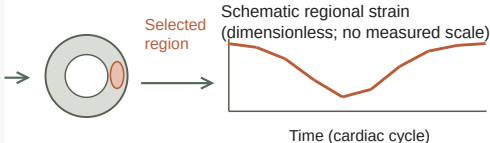
reference region / layer aggregation

measure derivatives

axes time summary

d Regional report

Conditional output given the above choices



Regional strain values are conditional on the inferred mapping and on the specified reference, measure, axes, region/layer, derivatives, time summary, and aggregation.